

Exploratory Data Analysis

Superstore Dataset Report

Comprehensive analysis of sales, profitability, discounts, category performance, regional differences, and time trends.

1. Dataset Overview & Data Preparation

9,994

RECORDS

21

COLUMNS

0

MISSING

0

DUPLICATES

Data Preparation Steps

- Dates:** Order and Ship Dates converted to datetime format.
- Derived Features:** Year, Month, and Shipping Days extracted.

Sales & Profit Distribution

Sales Distribution: Exhibits a strong right-skew, heavily concentrated in low-value transactions with prominent high-value outliers.

Profit Distribution: Highly skewed (skewness ≈ 7.56). Ranges from $-\$6,599$ to $\$8,399$ with notable positive and negative extremes.

Insight: Extreme values represent legitimate large-scale transactions rather than errors.

2. Key Relationships & Category Performance

Variable Relationships

Metric Pair	Corr.	Key Takeaway
Sales vs. Profit	0.48	Moderate positive relation
Discount vs. Profit	-0.22	Weak negative trade-off

Warning: Higher revenue/sales do not automatically guarantee profitability.

Category Performance Breakdown

Category	Sales Volume	Net Profit
Technology	~\$836K	~\$145K
Furniture	~\$742K	~\$18K
Office Supplies	~\$719K	~\$122K

Concern: Furniture generates massive Sales (~\$742K) but disproportionately low Profit (~\$18K).

3. Regional Performance & Correlation Matrix

Regional Analysis

Region	Sales	Profit
West	~\$725K	~\$108K (Strongest)
East	~\$678K	~\$92K
Central	~\$501K	~\$40K (Low Margin)
South	~\$392K	~\$47K

Note: Central generates higher volume than South but yields weaker profitability.

Full Correlation Summary

Pair	Corr.	Type
Sales & Profit	0.48	Moderate (+)
Sales & Qty	0.20	Weak (+)
Qty & Profit	0.07	Neutral
Discount & Sales	-0.03	Negligible
Discount & Profit	-0.22	Weak (-)

4. Key Findings, Trends & Business Recommendations

Key Findings & Trends

1. Sales distribution is right-skewed with low-value dominance.
2. Profitability exhibits high volatility with margin erosion periods.
3. Technology leads overall in both revenue and net margins.
4. Furniture suffers from low margins despite strong top-line sales.
5. Discounts show a detrimental impact on net profitability (-0.22).
6. Performance fluctuates consistently across 2014–2017.

Actionable Business Recommendations

- **Furniture Review:** Audit discount thresholds and product-level costs within the Furniture category.
- **Discount Control:** Impose stricter discount limits to protect profit margins from heavy erosion.
- **Central Region Focus:** Investigate operational inefficiencies driving low profitability in the Central region.
- **Dual-Metric Tracking:** Evaluate business units using both Sales and Profit simultaneously.

Final Conclusion

While top-line Sales are vital for market presence, they do not guarantee profitability. Technology and the West region drive sustainable growth, whereas Furniture and the Central region require immediate strategic intervention regarding discount policies and cost structures.

Recommendation Summary: Transition from pure revenue-focused targets to balanced profitability and margin-driven metrics across all product categories and regions.